



CROSS-MORPHOLOGY LOCOMOTION VIA
VISUAL LATENT ACTION WORLD MODELS

MISS DISTHORN SUTTAWET

A THESIS SUBMITTED IN PARTIAL FULFILLMENT
OF THE REQUIREMENTS FOR
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บทคัดย่อ

การพัฒนาหุ่นยนต์เดินด้วยการเรียนรู้แบบเสริมกำลัง (Reinforcement Learning) ในปัจจุบันมีข้อจำกัดสำคัญ คือ นโยบายควบคุม (policy) ที่ฝึกได้จะผูกติดกับสัณฐานวิทยา (morphology) ของหุ่นยนต์ตัวนั้นโดยตรง เมื่อรูปร่างของหุ่นยนต์ เปลี่ยนไป เช่น ความยาวขาเปลี่ยน นโยบายเดิมจะใช้ไม่ได้และต้องฝึกใหม่ทั้งหมด ซึ่งใช้เวลาและทรัพยากรสูง งานวิจัยนี้จึงศึกษา การเรียนรู้การกระทำแฝงที่ไม่ขึ้นกับสัณฐานวิทยา (morphology-agnostic latent action) จากวิดีโอในสภาพแวดล้อมจำลอง โดยประยุกต์แนวคิดของแบบจำลองโลกเชิงการกระทำแฝง (Latent Action World Model) ซึ่งเดิมใช้ในงานหีบจับวัตถุ มาสู่โดเมนการเคลื่อนที่ (locomotion) โดยงานวิจัยใช้แมลงกิ้งมั้ง (stick insect, *Medauroidea extradentata*) ในโปรแกรมจำลอง CoppeliaSim จำนวนสามสัณฐานที่มีความยาวขาต่างกัน (1.0, 0.75 และ 0.5 เท่า) โดยใช้ตัวเข้ารหัสภาพ V-JEPA2 แบบตรึงค่า (frozen) ร่วมกับแบบจำลองการเปลี่ยนผ่านเชิงผกผัน (Inverse Transition Model) และเชิงไปข้างหน้า (Forward Transition Model) เพื่อสกัดการกระทำแฝง z_t วัตถุประสงค์หลักคือเพื่อพิสูจน์ว่า z_t เกาะกลุ่มตามพฤติกรรม (เช่น การเดิน) แทนที่จะเกาะ กลุ่มตามรูปร่างขา และเพื่อแสดงว่าแบบจำลองโลกที่ผ่านการฝึกล่วงหน้าช่วยให้สัณฐานที่ไม่เคยเห็นมาก่อนเรียนรู้ได้เร็วขึ้นโดย ใช้ข้อมูลน้อยลง

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Abstract

Reinforcement-learning controllers for legged locomotion are tightly coupled to the body they were trained on: when leg length changes, the policy becomes unusable and must be retrained from scratch, at high cost in time and compute. This research studies whether a **morphology-agnostic latent action** can be learned from simulation video, adapting the Latent Action World Model paradigm, previously demonstrated only for robotic manipulation, to **legged locomotion**. The study uses a simulated stick insect (*Medauroidea extradentata*) in CoppeliaSim, instantiated as three morphologies differing only in leg length (1.0 $\times$, 0.75 $\times$, 0.5 $\times$). A frozen V-JEPA2 encoder is combined with an Inverse Transition Model and a Forward Transition Model to extract a latent action z_t . The objectives are (i) to demonstrate that z_t organizes by *behaviour* (e.g. walking) rather than by *body shape*, and (ii) to show that a world model pretrained on two morphologies lets an unseen third be learned with substantially fewer episodes than training from scratch. Crucially, the model is **never told its leg length**, in contrast to methods that read it from CAD, so the study tests whether the abstraction can be *learned* rather than supplied. Training grounds z_t on joint commands the simulator logs automatically, and that decoding module is retained as the only component that converts a latent action back into an executable, body-specific joint command.

Keywords: cross-morphology locomotion / latent action / world model / transfer learning / V-JEPA2

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LIST OF SYMBOLS

o_t	RGB observation (frame) at time t
e_t	Visual embedding of frame t (frozen encoder output)
z_t	Latent action inferred from the transition $e_t \rightarrow e_{t+1}$
a_t	Native joint command (18-D joint-position target)
$\hat{e}_{t+1}$	Predicted next embedding (Forward Transition Model output)
$\hat{a}_t$	Predicted joint command (Motion Decoder output)
λ	Loss-weighting coefficient
δ	Cliff's delta (effect-size measure)

LIST OF ABBREVIATIONS

RL	Reinforcement Learning
WM	World Model
CM-VLAWM	Cross-Morphology Visual Latent Action World Model (this work)
LAC-WM	Latent Action-Conditioned World Model
ITM	Inverse Transition Model
FTM	Forward Transition Model
MD	Motion Decoder
IK	Inverse Kinematics
CPG	Central Pattern Generator
JEPA	Joint-Embedding Predictive Architecture
AEP/PEP	Anterior / Posterior Extreme Position

CHAPTER 1 INTRODUCTION

1.1 Background and Significance

Legged robots are typically controlled by policies learned through reinforcement learning (RL). A persistent and expensive limitation of this approach is **morphology specificity**: a policy learned for one body does not generalize to another. If a robot’s legs are lengthened or shortened, its mass distribution changes, or a limb is damaged, the previously trained policy fails and a new one must be learned from scratch, a process that can take hours to days of training per body. In contrast, biological organisms share locomotion principles across vastly different body plans, suggesting that a shared, body-independent representation of *movement* is possible.

One promising direction is the **World Model**, an internal learned model of environment dynamics that predicts future states from actions and lets an agent “imagine” rollouts instead of always acting in the real environment (e.g. DreamerV3). A recent advance, the **Latent Action-Conditioned Robot World Model** (LAC-WM, Huang et al., 2026), learns a *unified latent action space* shared across multiple robot embodiments, and shows that a learned latent action generalizes across embodiments better than raw, body-specific action commands. However, LAC-WM was demonstrated only for **manipulation** (robot arms and hands), and, like other cross-embodiment methods, the embodiments it unifies have *structurally different action spaces*.

This research asks whether the same idea transfers to **legged locomotion**, and specifically to a setting the prior work does not address: bodies that share an *identical* action space and differ only in **limb length**. Here the challenge is not to reconcile different action dimensionalities, but to learn a latent action that captures the **behaviour** (the intent of a movement, such as “step forward”) independently of how a given body realizes that behaviour, because identical joint commands produce *different resulting motion* on a short leg versus a long leg. We ground this study in a simulated stick insect (*Medauroidea extradentata*), a well-studied model organism for insect walking, instantiated at three leg lengths.

The significance is twofold. Scientifically, it tests whether a morphology-invariant notion of behaviour can be learned from visual observation without the model ever being told the body’s dimensions. Practically, if such a representation transfers, it would cut the cost of deploying a new robot morphology by reusing locomotion skill already acquired instead of retraining from zero. We refer to this vision-based, cross-morphology approach as the **Cross-Morphology Visual Latent Action World Model (CM-VLAWM)**.

1.2 Objectives

1. To learn a morphology-agnostic latent action z_t from simulation video of a legged robot, by adapting the Latent Action World Model pipeline (frozen visual encoder $\rightarrow$ Inverse Transition Model $\rightarrow$ Forward Transition Model $\rightarrow$ Motion Decoder) from manipulation to locomotion.
2. To demonstrate quantitatively that z_t organizes by **behaviour** rather than by **morphology**, using linear-probe accuracy and cluster-separation (silhouette / variance-decomposition) metrics across three leg-length variants of the same body.
3. To demonstrate that a world model pretrained on two morphologies enables a third, unseen morphology to reach the same reconstruction quality with **significantly fewer training episodes** than a model trained from scratch (i.e. improved sample efficiency of transfer).

1.3 Expected Benefits

1. A demonstration that latent-action world models extend beyond manipulation to legged locomotion.
2. Evidence on whether morphology-invariant behaviour structure can be learned **without** privileged morphology information (no CAD/spec input), using only visual observation and auto-logged action labels.
3. A reusable evaluation methodology (behaviour-vs-morphology latent probing) and a controlled, single-axis leg-length benchmark for future cross-morphology locomotion studies.
4. A foundation for reducing the retraining cost of deploying legged robots of new body proportions.

1.4 Scope of the Research

1. The study is conducted **entirely in simulation** (CoppeliaSim v4.10 with the Bullet physics engine). No physical-robot or sim-to-real deployment is in scope.
2. The robot is a single species, the stick insect *Medauroidea extradentata*, instantiated as **three morphologies** that differ **only in leg length** (1.0 $\times$ baseline, 0.75 $\times$, 0.5 $\times$); all other properties (topology, degrees of freedom, 18-dimensional joint-position action space) are held identical. Training uses the 1.0 $\times$ and 0.5 $\times$ variants; the 0.75 $\times$ variant is held out for the transfer test. This is an **interpolation** study (the unseen body lies between the trained ones), not extrapolation.
3. The behaviours studied are **walking, turning, and stopping**. All three are generated through the inverse-kinematics data pipeline (Section 3.5.3) as task-space foot trajectories derived from an existing expert dataset (Larsen et al., 2023), at no architectural cost.
4. The visual encoder (V-JEPA2) is used **frozen**; training it is out of scope except for a possible fine-tuning of its final blocks if a domain gap is found.

Explicitly not claimed. To set expectations precisely, the following are outside the claims of this thesis: real-robot or animal transfer (the study is entirely in simulation); extrapolation beyond the trained leg-length range (the held-out body is an interpolation, between the trained ones); invariance to camera viewpoint (a single fixed viewpoint is used throughout); generalization to manipulation or complex terrain; and fully autonomous closed-loop control through the latent action. A further honest caveat: the inverse-kinematics data redesign (Section 3.5.3) creates comparable task-space objectives across bodies, but it does not guarantee identical contact dynamics or identical behaviour across morphologies, and the preliminary numbers of Section 3.7 are diagnostic and will be regenerated on the re-designed dataset.

1.5 Research Procedure

The project follows a staged, milestone-gated procedure in which each stage must pass before the next begins (this component-by-component validation strategy was set by the lab advisor):

1. **Morphology Gap Check:** confirm that the three leg-length variants produce measurably different behaviour under identical commands (otherwise the transfer question is vacuous).
2. **Visual Encoder Sanity Check:** confirm that the frozen V-JEPA2 features carry the behaviour-relevant signal (gait/contact state) needed downstream, before any training.
3. **World-Model Training:** train the Inverse/Forward Transition Models and Motion Decoder on the two training morphologies.
4. **Latent-Space Validation:** verify that the learned latent action z_t clusters by behaviour, not morphology.
5. **Transfer Evaluation:** measure the sample-efficiency gain of the pretrained world model on the unseen morphology, against a from-scratch baseline and a raw-action baseline.

CHAPTER 2 LITERATURE REVIEW

The review begins with a short, self-contained account of the fundamental concepts the rest of the chapter uses (Section 2.1), so that the work is understandable without prior background. It then draws on four bodies of work: world models, latent action models, cross-morphology legged control, and the biomechanics of insect walking (which motivates the model organism and the evaluation). Each is reviewed in turn, and the review closes with the specific research gap this work addresses and an argument for its central design choice.

2.1 Fundamental Concepts

This section defines, in plain terms, the concepts the rest of the review and the methodology rely on. It is limited to what this work actually uses; a reader familiar with machine learning can skip to Section 2.2.

2.1.1 Reinforcement learning and policies

Reinforcement learning (RL) is a framework in which an **agent** interacts with an **environment** over time. At each step the agent observes the current **state** (for a robot, its configuration and surroundings), chooses an **action** (for a legged robot, the target angles for its joint motors), and the environment responds with a new state and a scalar **reward** signalling how good the outcome was. The agent’s goal is to maximise the total reward accumulated over time. The rule the agent uses to choose an action from a state is called its **policy**, written $\pi : \text{state} \rightarrow \text{action}$. Learning a good policy by trial and error is how most modern legged robots are controlled.

The central difficulty this thesis addresses follows directly from this definition. A policy is a function learned for one specific body: it maps *that body’s* states to *that body’s* motor commands, and in the process it silently absorbs that body’s geometry and dynamics. When the body changes (a leg is lengthened), the same policy applied to the new body produces the wrong motion, because the state-to-action mapping it learned no longer matches the physics. This is why a policy is said to be “tied to its morphology,” and why a change of body normally forces retraining from scratch.

2.1.2 Neural networks, encoders, and embeddings

A **neural network** is a function with many adjustable numbers (parameters, or “weights”) that is *trained* by adjusting those numbers so the function’s output matches desired targets on many examples. Given enough data, neural networks can learn very complex mappings, such as from a raw image to a useful description of its content.

An **encoder** is a neural network that compresses a high-dimensional input (such as a $256 \times 256 \times 3$ image, which is almost two hundred thousand numbers) into a much smaller vector of numbers that summarises the input’s meaningful content. That output vector is called an **embedding** (in this work, e_t). Embeddings are useful because downstream models can operate on the compact, informative embedding instead of on raw pixels. When an encoder’s weights are held fixed and not adjusted during later training, it is called **frozen**; a frozen encoder acts as a fixed “translation” from images to embeddings.

2.1.3 Self-supervised learning

Neural networks are commonly trained by **supervised learning**, in which every training example carries a human-provided label (this image is a cat) and the network learns to reproduce the labels. Labels are expensive, so a second paradigm, **self-supervised learning**, creates the training signal from the data itself with no human labels: part of the input is hidden and the network is trained to predict it from the rest. Because the “answer” is just another part of the same data, self-supervised learning can use enormous unlabeled datasets. The visual encoder used here (V-JEPA 2, Section 2.2.2) is trained this way on about one million hours of unlabeled video, which is why it can be reused without any task-specific labels.

2.1.4 The Transformer and the attention mechanism

The learned models in this work (the encoder and the transition models) are built on the **Transformer** architecture, whose core is the **attention mechanism**. Attention is worth understanding because it explains how these models decide *what part of the input to look at*.

The intuition is as follows. Split the input into small pieces called **tokens** (for an image, each token is a small square patch of pixels; for a pair of frames, the tokens are all the patches of both frames). On its own, a single patch of a leg knows nothing about the rest of the scene. Attention lets every token **look at every other token and take a weighted summary of them**, so that each token’s representation becomes aware of its context. The patch of a foot can incorporate information from the patch of ground beneath it, or from where the same foot was in the previous frame.

Mechanically, each token emits three vectors: a **query** (“what am I looking for?”), a **key** (“what do I offer?”), and a **value** (“what content do I carry?”). To update one token, the model compares that token’s query against every other token’s key to get a set of relevance scores, converts those scores into weights that sum to one, and returns the weighted average of all the tokens’ values. A token thus pulls in most strongly from the tokens most relevant to it. When the tokens attend among themselves this is **self-attention**; when one set of tokens (for example a single learned query summarising “what changed”) attends over another set (for example all the patches of two frames) it is **cross-attention**. **Causal** self-attention simply restricts each token to attend only to earlier positions, which is a natural

way to express “what happened between an earlier and a later frame.” Stacking many such attention layers, interleaved with small per-token networks, is what makes a Transformer, and it is the mechanism by which this work’s Inverse and Forward Transition Models (Section 3.4) decide which moving parts of the scene define the latent action.

2.1.5 Evaluation: probing and metrics

Because the central claim of this thesis is about *what information a representation contains*, the evaluation uses tools that measure exactly that.

A **linear probe** is the primary tool. To ask “does embedding e_t contain information about property Y (for example, which feet are planted)?”, one trains a *simple linear classifier* to predict Y from e_t and measures how well it does. The classifier is deliberately kept simple (linear) so that success reflects the *quality of the embedding* rather than the cleverness of the classifier: if even a linear model can read Y out of e_t , then Y is present in an accessible form. Probing therefore measures the **presence** of information.

Several standard measures accompany the probe. **Accuracy** is the fraction of predictions that are correct, but it is misleading when the classes are imbalanced (a model that always predicts the most common class can score high while learning nothing). **Macro-F1** corrects for this by scoring each class separately and averaging with equal weight, so a model that ignores rare classes is penalised; it is the metric used here for the imbalanced foot-contact labels. Every probe is compared against a **chance baseline** (the score a random guesser would obtain, for example 1/8 for eight equally likely classes) so that “above chance” is well defined. To ensure a probe has genuinely learned rather than memorised, results are reported under **cross-validation** (the data is split into folds; the model is trained on some folds and tested on the held-out fold, repeated so every example is tested once) and against a **shuffled-label control** (the labels are randomly permuted; a model that still scores above chance would be exploiting an artefact, so a control that collapses to chance confirms the real result is genuine). Finally, a probe measures whether information is *present*, but not whether it *dominates* the representation; for that this work also reports the **silhouette score**, a measure of how cleanly the embeddings separate into groups, which captures whether a property is the main axis of variation rather than merely decodable. Section 3.6.1 explains why both are needed.

2.2 World Models and Video Representation Learning

2.2.1 What a world model is, and why it is used here

In reinforcement learning an agent must decide which action to take in each state. A *model-free* agent learns this mapping by trial and error directly in the environment, which is data-hungry and, for a physical robot, slow and risky. A *world model* takes a different route: it first learns to **predict the consequences of actions** (given the current state and an action, it predicts the next state) and thereby builds an internal, learned simulator of the

environment’s dynamics. Once such a model exists, the agent can train its policy by “imagining” long sequences of hypothetical actions inside the model, without touching the real environment. This is the sense in which world models let an agent “learn in imagination.”

The reference implementation is DreamerV3 (Hafner et al., 2023). Rather than predicting raw future pixels, which is expensive and forces the model to waste capacity on visually irrelevant detail, DreamerV3 encodes each observation into a **compact latent state** and learns the dynamics *between latent states*. A single agent with one fixed hyperparameter setting was shown to master more than 150 distinct tasks, from control benchmarks to collecting diamonds in Minecraft from raw pixels, which established learned latent-state dynamics as a robust and general tool. Two properties of this design matter for the present work. First, because the policy is trained against the model’s predictions, the *representation* the model learns is the object the policy actually consumes; if that representation can be made body-independent, everything built on top of it inherits that property. Second, and central to this thesis, a world model must answer the question “given this state and this action, what happens next?”, and that predictive objective is an operational *test* of whether a proposed action representation captures meaningful change, independent of any downstream control task. This project uses a world model less to imagine rollouts than to use its transition-prediction objective as such a test.

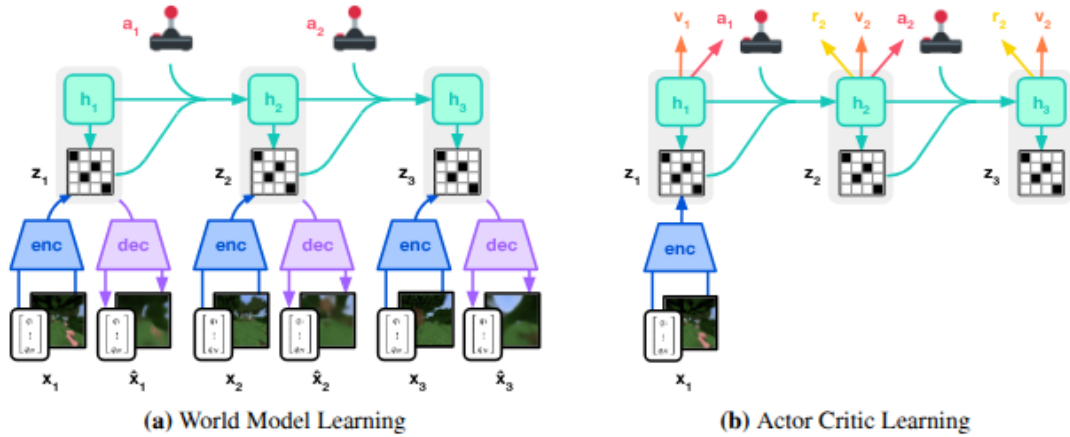


Figure 2.1 The DreamerV3 world model and actor-critic (Hafner et al., 2023). Observations are encoded into a compact latent state whose evolution a sequence model predicts from actions; the actor and critic then learn on trajectories *imagined* inside the model rather than in the real environment. Predictions are conditioned on the agent’s *native* action, the body-specific conditioning signal this thesis replaces with a latent action.

A limitation of the standard formulation, which motivates Section 2.3, is that DreamerV3 and its relatives learn a separate model per domain and condition their predictions on the agent’s **native action**, the robot’s own motor command. Nothing is shared across different bodies, and the native command is assumed to be the meaningful conditioning signal. When the same command has a different physical meaning on a different body, that assumption breaks.

2.2.2 Video foundation models and the JEPA objective

To apply a world model to visual observations, one first needs to turn images into useful features. A *video foundation model* is a large network pretrained on massive unlabeled video so that its features transfer to many downstream tasks. This work uses V-JEPA 2 (Assran et al., 2025). Its training objective is the distinguishing feature: instead of reconstructing masked pixels (as in masked autoencoders), a JEPA (Joint-Embedding Predictive Architecture) masks part of the input and predicts the **representation** of the masked part rather than its pixels. Predicting in representation space frees the model from modelling unpredictable low-level detail (exact textures, lighting noise) and pushes it toward features that capture *how things move and change*. Trained by this objective on roughly one million hours of internet video, V-JEPA 2 learns motion-relevant features that transfer to action recognition, prediction, and planning, and reports state-of-the-art results on motion-centric benchmarks while remaining competitive on appearance-centric ones.

Two aspects of V-JEPA 2 are directly relevant. First, the accompanying V-JEPA 2-AC (action-conditioned) variant uses the pretrained encoder as a **frozen per-frame image encoder** feeding a lightweight downstream predictor, without fine-tuning the encoder. This is the precedent for how the encoder is used here, and it is why a large frozen backbone can be treated as a fixed perceptual front-end (Section 3.3). Second, V-JEPA 2 was trained on real-world video only, with no rendered or simulated data. Its behaviour under a rendering domain gap is therefore not characterised by the original work, which is one reason this project verifies empirically (Section 3.7.2) that the frozen features carry the required locomotion signal on simulated stick-insect footage before relying on them. Figure 2.2 summarises the two training stages.

2.3 Latent Action Models and Latent Action World Models

2.3.1 Inverse and forward dynamics models: the general building block

Two complementary models recur throughout control, world modelling, and representation learning, and both the latent-action idea and this project’s architecture are built from them. A **forward dynamics model** answers “given the current state and an action, what is the next state?” and maps (s_t, a_t) to $\hat{s}_{t+1}$. A world model’s transition predictor is exactly a forward dynamics model, and it is the component an agent rolls forward to imagine the future (Section 2.2.1). An **inverse dynamics model** answers the opposite question, “given two consecutive states, what action caused the transition?” and maps (s_t, s_{t+1}) to $\hat{a}_t$.

Both are established, general-purpose techniques rather than the invention of any single paper. Inverse dynamics models have long been used to recover actions from observation alone: the Intrinsic Curiosity Module (Pathak et al., 2017) pairs an inverse and a forward model so that an agent can explore using its own prediction error, and Video PreTraining (Baker et al., 2022) trained an inverse dynamics model to *label* large quanti-

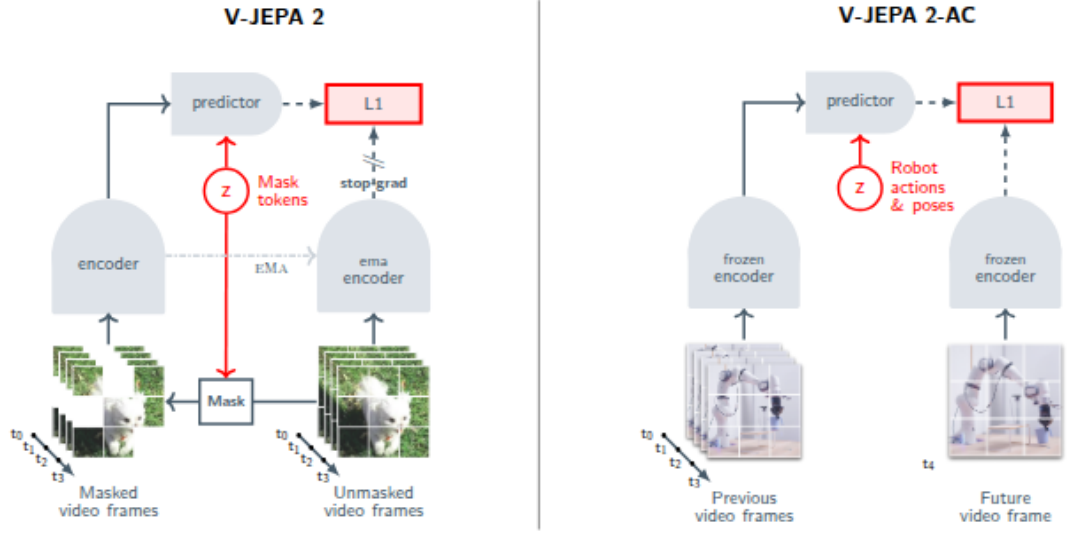


Figure 2.2 V-JEPA 2 multistage training (Assran et al., 2025). Left: self-supervised pre-training on internet-scale video with a mask-prediction objective in *representation* space (the JEPA objective), which favours motion-relevant features. Right: the encoder is then frozen and an action-conditioned predictor (V-JEPA 2-AC) is learned on top. This project reuses only the frozen encoder as a per-frame visual front-end (Section 3.3).

ties of unlabeled gameplay video with the actions that must have produced them, so that behaviour could then be learned from video without action annotations. Forward dynamics models are the core of the world-model line through DreamerV3.

The two models are naturally combined. If the “action” recovered by the inverse model is not the robot’s native command but an inferred latent code, and the forward model is asked to predict the next state from the current state plus that latent, then the pair *jointly learns* a latent action: the inverse model proposes it, and the forward model’s prediction error tests whether it is sufficient to explain the transition. This inverse-forward pairing is the shared skeleton of the latent-action methods reviewed next (Section 2.3.2) and of the base method this project adapts (Section 2.3.3). (*This project instantiates the two models on visual embeddings and, for that reason, refers to them as the Inverse and Forward Transition Model; the naming rationale is given in Section 3.4.*)

2.3.2 The idea of a latent action

The conditioning problem raised at the end of Section 2.2.1 is that the native motor command is a poor shared signal across bodies. A *latent action* addresses this by not using the native command at all. The idea is to look at two consecutive observations and infer a compact code (the latent action) that explains the change between them. Formally, an *inverse* model reads (o_t, o_{t+1}) and outputs a latent action z_t ; this is analogous to inverse dynamics in control, which asks “what action must have produced this transition?” The key property is that z_t is defined by *observed change*, not by any robot’s motor format, so a single latent space can describe several embodiments at once, and it can be learned from video that has **no action labels** at all.

This places the approach within the broader paradigm of **learning from observation** (also called imitation from observation): learning behaviour by watching, without access to the demonstrator’s action labels, and, in the strongest form, from a third-person external view of a body that is not the learner’s own. The latent-action line is one way to make observation-only learning tractable, by recovering a usable action interface from the video itself.

Several lines of work established this. Genie (Bruce et al., 2024) learned controllable, playable environments from large collections of unlabeled internet video by inferring a discrete latent action between frames, showing that a usable action interface can emerge without any action supervision. LAPA (Ye et al., 2024) carried latent-action pretraining toward real robot control, and CLAM (Liang et al., 2025) introduced *continuous* latent actions, which are the closest match to the continuous joint control needed for locomotion (a discrete code is ill-suited to smooth, graded limb movement). Zhang et al. (2025) provide a theoretical analysis of what latent action models actually capture, which is useful when interpreting whether a learned latent encodes behaviour or merely copies the next observation, a failure mode this project must actively prevent (Section 3.1, cross-augmentation).

Most directly related to the cross-embodiment goal, UniSkill (Kim et al., 2025) learns an **embodiment-agnostic skill representation** from video (including human video) so that a robot can imitate a demonstration performed by a different body. Architecturally it is again the same building block of Section 2.3.1: it uses an **inverse skill dynamics** model (infer a skill code from an observation pair) and a **forward skill dynamics** model (predict a future observation from the current one plus the skill), confirming that the inverse/forward pairing is the standard mechanism across this literature. UniSkill matters here for two reasons. First, it is evidence that a body-independent representation of *skill* can, in fact, be learned from observation, which supports the feasibility of the present question. Second, and importantly, it differs from this project in *how* the invariance is obtained: UniSkill is trained on cross-embodiment data and explicitly optimises its representation to align across bodies, whereas this project asks whether morphology-invariance can **emerge** from a world-model bottleneck *without* any dedicated cross-embodiment alignment objective and without the morphology ever being supplied. UniSkill also targets human-to-robot manipulation and imitation, not a controlled single-axis leg-length study or a dynamics-prediction world model. The contrast, invariance *aligned by design* versus invariance *tested for emergence*, is what sharpens the contribution of this work. An explicit alignment objective of this kind is a possible future extension rather than a contingency the primary result depends on: if morphology-invariance does not emerge, that outcome is itself a valid and informative result about the limits of the emergent approach in this setting. Figure 2.3 shows its structure.

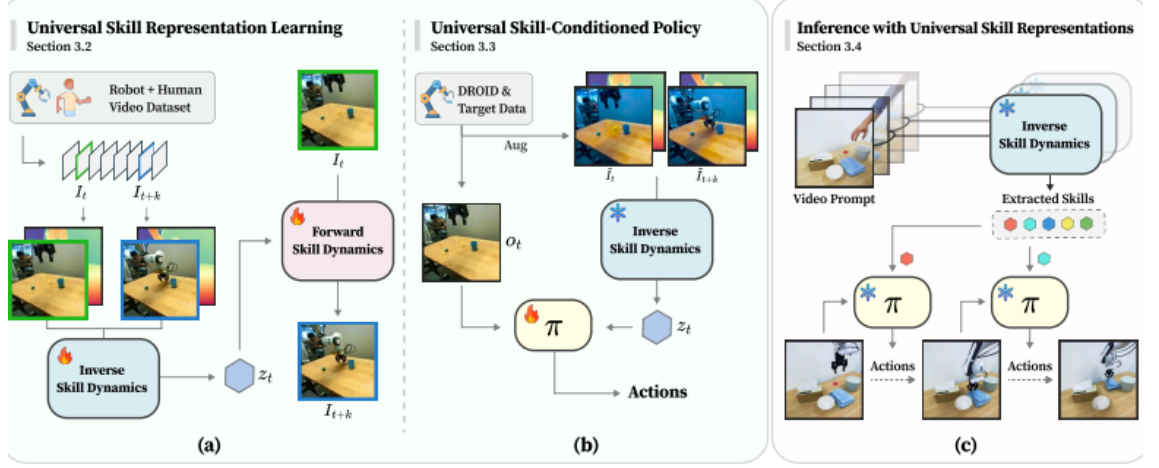


Figure 2.3 Overview of UniSkill (Kim et al., 2025). (a) An Inverse Skill Dynamics model and a Forward Skill Dynamics model are jointly trained on diverse video to encode a body-independent skill representation; (b) a skill-conditioned policy is trained on robot data; (c) skills extracted from a human-video prompt are executed on the robot to reproduce the behaviour. The inverse/forward pairing matches this project’s ITM/FTM (Section 3.4); the difference is that UniSkill *aligns* its representation across bodies by training on cross-embodiment data, whereas this project tests whether such invariance *emerges* without any alignment objective.

2.3.3 Latent Action World Models (the base method)

LAC-WM (Huang et al., 2026) combines the latent-action idea with a world model to unify a single latent action space across *heterogeneous* manipulation embodiments (for example a robot gripper, a bimanual humanoid, and human hand keypoints, whose action spaces have different dimensions and meanings). Its architecture is the pipeline this project adapts, and it has four parts. (i) A **frozen visual encoder** (V-JEPA 2) turns each frame into an embedding e_t . (ii) An **Inverse Transition Model (ITM)** reads the pair (e_t, e_{t+1}) and produces the latent action z_t , meaning “what happened between these two frames.” (iii) A **Forward Transition Model (FTM)** takes (e_t, z_t) and predicts the next embedding $\hat{e}_{t+1}$; its reconstruction error tests whether z_t , together with the current state, is sufficient to explain the observed transition. (iv) A **Motion Decoder (MD)** maps z_t back to the ground-truth motor action, which anchors the latent to real, executable motion so that it cannot collapse into a trivial or meaningless code. The world model is then conditioned on z_t rather than on any robot’s native command, so one latent space covers several embodiments, and, importantly, adding embodiments *improves* the shared model rather than fragmenting it. This pipeline is shown in Figure 2.4.

Because z_t is partly supervised to help reconstruct e_{t+1} , the ITM could cheat by copying the next frame’s content directly into z_t instead of learning the abstract action. LAC-WM prevents this with **cross-augmentation**: two independent augmentations of the frame pair are encoded, the ITM sees one while the reconstruction target is drawn from the other, so a copied pixel-level representation no longer matches the target and z_t is forced to capture augmentation-invariant change. This project inherits the mechanism.

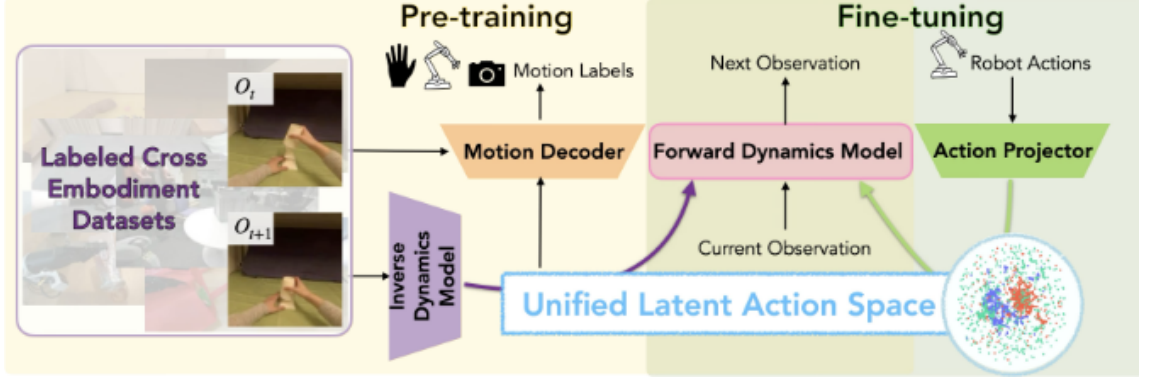


Figure 2.4 LAC-WM architecture (Huang et al., 2026), the pipeline this thesis adapts. During pretraining an inverse model encodes an observation pair into a unified latent action that conditions a forward model, and a motion decoder maps the latent back to each embodiment’s motor labels; a finetuning-stage action projector lets the world model also condition on an unseen body’s raw actions. The paper’s inverse/forward *dynamics* models are what this project instantiates on visual embeddings and renames the Inverse/Forward *Transition Model* (Section 3.4).

LAC-WM reports two headline results: its unified latent action transfers across embodiments better than an explicit-action baseline (EAC-WM, which is forced to use per-embodiment action encoders), and its performance *scales positively* with embodiment diversity. Its motivating premise, however, is heterogeneity of *action spaces*: the baseline pathology it improves on exists precisely because different manipulators have genuinely different action formats. That premise does **not** hold in this project’s single-species, leg-length-only setting, where all three bodies share the same 18-dimensional action format. Why a latent action is nonetheless needed (because the same behaviour is realised by different commands on different bodies) is taken up in Section 2.6.

2.4 Cross-Morphology and Cross-Embodiment Legged Locomotion

Making one controller work across bodies is an active problem, and the existing strategies differ mainly in *what information about the new body they consume*. This subsection reviews the strategies most relevant to legged locomotion; Section 2.7.3 organises them by the single question of where that information must come from.

2.4.1 Per-body retraining

The baseline strategy is simply to train a new controller for each body. DreamerV3 (Section 2.2.1) is the representative here: it is highly general across *tasks* but learns a fresh model per domain, with nothing carried across bodies. This is the cost the field is trying to avoid, and it is the reference against which the sample-efficiency benefit of the present method is measured (Section 3.6.3).

2.4.2 Explicit morphology conditioning

QWM (Danesh et al., 2026) introduces a morphology-conditioned world model that generalises zero-shot to unseen quadruped morphologies, and it is the strongest cross-morphology transfer result in this area. It achieves this by reading physical parameters (limb lengths, masses, torque limits) out of each robot’s CAD/USD description and conditioning the world model on those numbers. Its observation channel is proprioception, and it does not learn a latent *action*: the morphology is supplied, not inferred. QWM is therefore the closest and strongest prior art on one axis (world-model transfer across quadruped bodies) and the clearest contrast on another (it assumes an accurate machine-readable description of the new body). Notably, QWM routes what it calls “unmodeled real-world residuals” through a separate dynamic latent, an implicit acknowledgement that the design file does not fully describe the physical body, a point this project develops in Section 2.7.3. Figure 2.5 shows its framework.

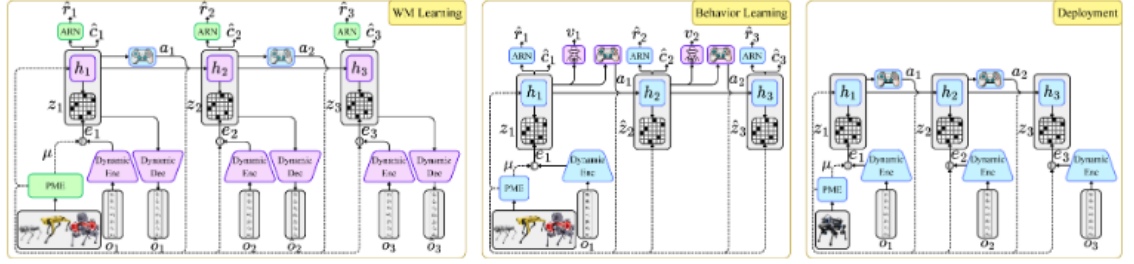


Figure 2.5 The QWM framework (Danesh et al., 2026). A Physical Morphology Encoder derives a static embedding from each robot’s USD design file and conditions the world model on it (left); a policy is then learned entirely in imagination for a new morphology (middle) and deployed by injecting that morphology’s embedding (right). The morphology is *supplied* from a design file, exactly the privileged input this thesis removes by inferring behaviour from external vision.

2.4.3 Online system identification

A related family infers hidden dynamics parameters online from a short window of recent interaction: the robot moves, observes its own responses, and estimates what body it is. This removes the need for a design file but requires access to the robot’s internal signals (proprioception, past commands) and enough new interaction to identify the parameters, so it still assumes an instrumented, controllable body.

2.4.4 Shared policy with per-robot adapters

L3P (Zheng et al., 2025) is a transferable latent-to-latent locomotion policy: a shared latent policy backbone is wrapped by a per-robot observation encoder and a per-robot latent-action decoder, so control transfers across diverse legged robots. This shares *structure* across bodies, but the per-robot encoder and decoder must be fitted from that robot’s proprioception and foot force, with its joint ordering and actuator conventions known. Its decomposition (Figure 2.6) is a close structural parallel to this project’s: a shared latent

policy with body-specific encoders/decoders mirrors this work’s morphology-agnostic latent action with a body-specific Motion Decoder. The difference is the channel and the direction of information flow: L3P fits per-robot adapters from the body’s *internal* proprioception, whereas this project infers the latent from *external* vision and never fits a per-body input encoder.

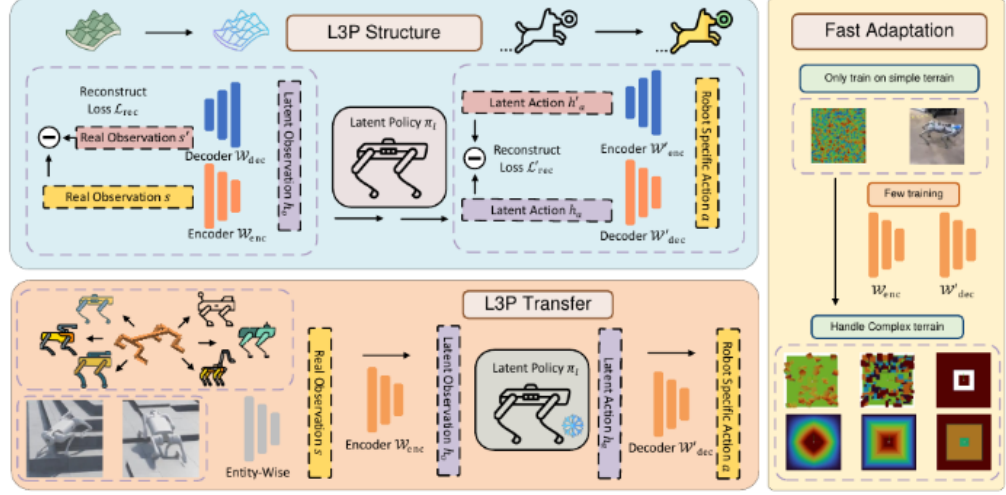


Figure 2.6 The L3P framework (Zheng et al., 2025). A shared latent policy backbone is wrapped by a per-robot observation encoder (raw sensing $\rightarrow$ shared latent space) and a per-robot latent-action decoder (latent action $\rightarrow$ robot-specific motor command), with recovery modules enforcing latent-space consistency. This shared-latent-with-body-specific-heads structure parallels this project’s latent action plus Motion Decoder, but L3P operates on proprioception and must fit an encoder and decoder per robot.

Graph- and transformer-based controllers such as MetaMorph (Gupta et al., 2022) learn morphology-agnostic policies over many procedurally varied bodies, and Ai et al. (2025) study embodiment scaling laws, showing that training across a large distribution of bodies improves generalisation. These establish cross-morphology control as tractable but approach it through policy transfer conditioned on the body’s internal representation, not through a vision-based latent action inferred from observed transitions.

2.4.5 Latent-action approaches in locomotion

Closest in spirit, Li et al. (2021) plan in a *learned latent action space* for legged locomotion and demonstrate the method on both a hexapod and a quadruped. Crucially, their framework is re-instantiated and trained *separately per robot*: no single latent space or dynamics model is shared or transferred across bodies, so it does not perform cross-morphology transfer in the sense studied here. It shows that latent actions are a natural fit for legged control, without yet making the latent space itself the object that transfers.

Across all five families, the observation channel is proprioception or a design file, and the body information is obtained from inside the body or its specification. None studies whether an **external, vision-based latent action inferred from observed transitions**, with morphology never supplied, can serve as the shared representation.

2.4.6 Synthesis: positioning the methods

Table 2.1 places the main reviewed methods on the axes that matter for cross-morphology transfer: what they *observe*, where they get *information about the new body*, how they represent an *action*, whether a single model *transfers across bodies*, and the domain in which they were demonstrated. Read down the “morphology information” column, every prior method obtains it from inside the body or from a design file; read across the bottom row, the present work is the only entry that observes externally (vision), is never given the morphology, uses an inferred latent action, and targets legged locomotion. The empty region this exposes (external observation, morphology never supplied, latent action, locomotion) is precisely the research gap of Section 2.6. It is also worth noting that latent-action and cross-embodiment-from-video methods (LAC-WM, UniSkill, and the latent-action pretraining line) are overwhelmingly demonstrated on **manipulation**; legged locomotion, where the same command produces different motion purely through body geometry, remains largely unaddressed by this family.

Table 2.1 Positioning of cross-morphology / cross-embodiment methods. “Transfers across bodies” asks whether a single trained model is reused on a new body without per-body retraining.

Method	Observation	Morphology information from	Action representation	Transfers across bodies?	Domain
DreamerV3 (Hafner 2023)	state / pixels	retrained per body	native command	no (per-body model)	general control
QWM (Danesh 2026)	proprioception	CAD / USD design file	native command	yes, zero-shot	quadruped
Online system ID	proprioception	inferred from interaction	native command	yes, after adaptation	legged / general
L3P (Zheng 2025)	proprioception + force	per-robot fitted adapter	shared policy latent	partial (adapter per body)	legged
MetaMorph (Gupta 2022)	proprioception + morphology graph	body graph / structure	native command	yes, within trained set	procedural bodies
Li et al. (2021)	proprioception	retrained per body	learned latent action	no (separate per body)	hexapod, quadruped
LAC-WM (Huang 2026)	vision	inferred from transition	learned latent action	yes	manipulation
UniSkill (Kim 2025)	vision	aligned by training objective	skill latent	yes	manipulation / human
This work	vision (external)	never supplied	learned latent action	under test	legged locomotion

2.5 Biomechanics of Insect Walking and Prior Work

2.5.1 Why a stick insect, and how it walks

The stick insect (*Carausius morosus*, *Medauroidea extradentata*) is a classical model organism for legged locomotion, studied for decades by Cruse, Bässler, Büschges, Dürri and colleagues. It is chosen here for three reasons: its walking is exceptionally well characterised, so behaviour labels can be defined against a solid reference; a validated simulation model of the exact species already exists (Larsen et al., 2023, Section 2.5.3); and, as a slow, statically-oriented walker, it exercises the cross-morphology question on a body whose gait is governed by leg coordination rather than by high-speed dynamics.

Coordination in stick insects arises not from a central clock but from **decentralised local rules** (the “Cruse rules”) acting between neighbouring legs (for example, a leg is discouraged from starting its return stroke while its neighbour is still returning, and a leg that completes a step encourages the next). A single walking cycle per leg alternates a *stance* phase (foot planted, propelling the body) with a *swing* phase (foot lifted, returning forward). The standard quantitative descriptors are the **anterior and posterior extreme positions (AEP/PEP)**, the boundaries of the stance stroke; the **duty factor**, the fraction of the cycle a foot is planted; and the **inter-leg phase**, the timing offset between legs. A well-known consequence of the decentralised control is that straight walking is largely an *emergent* property of the local rules and bilateral symmetry rather than the output of a dedicated heading controller: in classic experiments even a single remaining leg still produces the correct stepping pattern.

Insect gaits form a spectrum with walking speed: at high speed a *tripod* gait (two alternating sets of three legs) is used, while at low and medium speed the coordination is a more continuous *wave* or *tetrapod-like* pattern in which legs are recruited in a back-to-front sequence and the clean two-group structure of the tripod is not present. This spectrum matters for the present study for two practical reasons. First, it justifies labelling behaviour by the directly observable **foot-contact state** (which feet are planted) rather than by an abstract phase, because contact is a well-defined physical event across the whole gait spectrum. Second, it sets the correct expectation for the recorded data: the reference gait used here is a slow-walking pattern, and analysis of both the pilot recordings and the mature expert controller of Larsen et al. (2023) confirms a phase-staggered wave rather than a clean tripod (the two canonical tripod groups do not co-activate; the “clean tripod” state is essentially absent). This is the real gait of a slow-walking stick insect, consistent with the biology, and is not treated as a defect to be corrected.

2.5.2 Dynamic similarity and why leg length changes the outcome

That identical commands produce different locomotion on different-length legs (Section 3.7.1) is expected from biomechanics, not an artefact. The dynamic-similarity theory of gaits (Alexander and Jayes, 1983) formalises how gait and speed scale with size: geometrically similar bodies of different scale move similarly only when compared at matched dimensionless speed (the Froude number), and identical joint kinematics on a longer limb sweep the foot through a larger arc and change contact timing and ground reaction. The sub-linear scaling of walking speed with leg length observed in the pilot (an exponent near 0.69 rather than 1.0, Section 3.7.1) is consistent with this picture: the outcome of a fixed command is a function of the body’s geometry, which is exactly the dynamics heterogeneity the latent action is meant to abstract away.

2.5.3 Directly related prior work

Directly relevant, Larsen et al. (2023) model the *same species* in CoppeliaSim under a decentralised CPG (central pattern generator) controller with foot-contact feedback, explicitly handling heterogeneous leg lengths, and Chuthong et al. (2026) study resilient stick-insect-inspired control. These provide the biological data, the validated simulation model, and a controller lineage that this project builds upon. The mature logged rollouts of this controller family are the source of the shared foot trajectories and binary contact labels used in the dataset redesign (Section 3.5.3). The present project reuses this substrate but pursues a different research question: a learned, vision-based, morphology-invariant *representation* of locomotion, rather than a hand-designed controller.

2.6 Research Gap

2.6.1 The gap in the literature

Prior cross-morphology world models either (a) require **explicit** morphology specifications from CAD (QWM), (b) are trained **separately per body** without transfer (Li et al.), or (c) are demonstrated only for **manipulation** (LAC-WM). None studies whether a **vision-grounded latent action**, with morphology **never provided to the model**, organizes by *behaviour* rather than *body shape*, and none isolates the effect of a single morphological axis (leg length) on that latent. This project addresses exactly that gap: an implicit/spec-free, vision-based latent-action world model for locomotion, evaluated with a controlled single-axis leg-length sweep and an explicit behaviour-vs-morphology latent probe. We are careful **not** to claim vision-only learning: the latent is grounded during training on auto-logged joint commands (a by-product of simulation, not used at inference), following LAC-WM.

The study is conducted in simulation deliberately, not as a limitation of convenience. Answering the research question requires *ground-truth* control over the one variable under study: three bodies that are provably identical in every respect except leg length, driven by provably identical commands, with exact per-foot contact and body pose logged for evaluation. Only simulation provides this level of control and measurement, which is what makes the behaviour-versus-morphology comparison clean. Real-robot and sim-to-real transfer are explicitly out of scope (Section 1.4); the contribution is the controlled demonstration that the abstraction can, or cannot, be learned, which is a prerequisite to any later physical deployment.

2.6.2 Why a latent action is needed even though the action format is shared

LAC-WM’s shared latent is justified by its embodiments having genuinely *different action formats*: a gripper, a bimanual humanoid, and human-hand keypoints do not even have the same number of action dimensions, which forces its explicit-action baseline into a separate action encoder per embodiment (Section 2.3.3). That justification does not apply here. All three bodies in this study have the same six legs and eighteen joints, so their commands live in the same space, $\mathbb{R}^{18}$. A reviewer could reasonably ask why a latent action is needed at all when the command format is already common.

The answer is that a shared action *format* is not a shared action *meaning*. The same behaviour is produced by *different commands* on different bodies, and this project’s data design makes that difference explicit rather than incidental: the inverse-kinematics re-targeting of Section 3.5.3 takes one shared foot trajectory and solves it into a *different* joint command for each leg length. Section 3.7.1 shows the same fact from the opposite direction: a single fixed command produces non-overlapping motion across bodies ($p = 0.0079$, complete separation). In both directions the mapping between command and behaviour is body-dependent.

A latent action is therefore motivated not by reconciling different action dimensions but by *separating what is shared from what is body-specific*. The latent z_t is intended to carry the behaviour, which is common across bodies, while the Motion Decoder performs the body-specific conversion from that behaviour to a particular body’s joint command. This division is exactly what the architecture implements, and it is what makes the single-axis leg-length setting a clean instance of the cross-embodiment problem: every difference except leg length is held fixed.

2.6.3 How the value of the latent action is tested

The primary evidence is direct: the latent z_t is probed on its own for whether it *raises* cross-morphology behaviour transfer and *lowers* morphology decodability relative to the raw visual features, at the same time (Section 3.6.1). This examines z_t as a representation and needs only the held-out body’s video, so it does not depend on a comparison against the native command.

As a supporting check, the design also compares a latent-conditioned transition model against one conditioned on the body’s own (inverse-kinematics-retargeted) joint command, under an identical encoder, capacity, dataset, and budget, evaluated zero-shot on the held-out body (Section 3.6.3). Two points keep this honest and are stated in full there: reconstruction loss structurally favours the latent (it is inferred from the transition and is higher-dimensional than the command), so an observation-only baseline is reported to isolate the action’s real contribution; and the raw command is a *privileged* input, since obtaining the correct command for a new body requires that body’s kinematics, which the motivating scenario assumes is unavailable, whereas the latent is inferred from observation. If the latent adds nothing over the raw command even with that handicap, the thesis reports that honestly.

2.6.4 Why success requires a two-sided criterion

Framed abstractly, the goal is a representation that is *invariant* to a nuisance factor (morphology) while *preserving* a factor of interest (behaviour). This is the standard problem of learning an invariant, or partially disentangled, representation, and it comes with a standard pitfall that dictates how success must be measured. It is trivial to make morphology unrecoverable: a representation that discards all information (one that maps every input to the same constant) leaks nothing about the body, but it also leaks nothing about the behaviour and is therefore useless. A criterion that only rewards *reduced morphology decodability* would score this degenerate “collapse” as a perfect result. For this reason the success criterion is deliberately **two-sided**, requiring both that morphology becomes *less* decodable from z_t than from the raw features **and** that behaviour remains *more* transferable, measured at the same time (Section 3.6.1). Only a representation that removes the nuisance factor *while retaining the content* satisfies both, which is what makes the test meaningful rather than self-confirming. The same reasoning underlies treating the held-out medium body as a genuine test: it is a held-out *domain*, and reaching it requires that the representation generalise across the morphology axis rather than merely fit the two training bodies.

2.7 Why the Observation Modality Is Part of the Research Question

Using vision instead of proprioception or CAD parameters is not presented here as a technological preference. The modality determines whether the research question can be asked at all, and it changes what the eventual answer would mean.

2.7.1 Explicit morphology answers a different question

QWM supplies limb lengths, mass, and torque limits to the world model and shows that generalization follows. That establishes something worth knowing: *given* an accurate description of the body, a world model can compensate for it. It cannot establish whether a body-independent notion of behaviour is *learnable*, because the answer has already been provided in the input. Our question is the second one, and it is only well posed if the model is never told what body it is looking at. Spec-free observation is therefore a requirement of the experiment rather than a design choice, and vision is the natural spec-free channel.

2.7.2 Vision is the adversarial modality, not the convenient one

A reasonable objection is that vision is chosen because it is fashionable. Our own preliminary measurements argue the opposite. Morphology is decodable from the raw visual features at approximately 100 percent, with a morphology silhouette of +0.0835, and the leading principal component of the embedding orders the three bodies by leg length with no labels supplied, so body shape is a dominant axis of the visual representation. A necessary caution accompanies this figure: because each body is recorded in a single session, morphology and recording session are confounded in the pilot data, and the multi-session control that would attribute the signal specifically to leg length is planned rather than done; either way, this is exactly the signal the latent action must remove. The confound-maximising nature of vision is best stated through what the *command* carries rather than through proprioception in general. In this study’s control setup the joint *commands* are bit-identical across the three bodies, so the command itself carries no morphology signature; the resulting *measured* proprioception (foot forces, realised joint state) does differ across bodies, but the standard proprioceptive channel a controller conditions on would not, by construction, separate them in this setup. Vision, by contrast, sees the body shape directly and in full. Vision therefore maximises the confound that the latent action is supposed to remove, and demonstrating behaviour-level invariance under maximal visible morphological difference is a stronger result than demonstrating it where the confound is weak.

2.7.3 The requirements differ in where they land, not in whether they exist

It would be inaccurate to say that specification-based and proprioception-based methods have requirements and the present method does not. The present method requires an external camera with a view of the body, and it requires logged joint commands from the bodies used during pretraining. The relevant difference is that these requirements fall on the observer, whereas the alternatives place their requirements on the observed body itself.

A CAD or USD file describes the design, not the individual robot: wear, payload, manufacturing tolerance, and damage are absent from it, and these are exactly the conditions under which morphology adaptation is most needed. QWM implicitly concedes this by routing “unmodeled real-world residuals” through its dynamic latent. Proprioception, as used by L3P (Zheng et al., 2025), requires the target body to carry joint and force sensing and requires its joint ordering and conventions to be known, so the per-robot encoder and decoder can be fitted. Both preconditions can be satisfied only by a party with access to the interior of the body or to its design record. A camera can be positioned by anyone who can see the subject.

This matters because the bodies for which morphology adaptation is most valuable are frequently ones where interior access is unavailable in principle rather than by oversight: animals, which cannot be fitted with joint encoders; robots whose configuration has diverged from their documentation through repair, payload, or wear; hardware acquired from another party without published kinematics; and, in the limiting case, extinct animals reconstructed from skeletal remains and trackways, for which no specification exists and none can be produced. For these, an external view is not a preferred channel but the only remaining one.

2.7.4 Only visual data scales beyond bodies one already owns

Recordings of legged locomotion exist in very large quantity for animals and for robots built by other groups. Proprioceptive logs and CAD files do not exist for bodies one did not build and instrument. A latent action grounded in vision is the only version of this idea that could later ingest observations of embodiments outside the laboratory, which is the long-horizon reason the cross-embodiment literature cares about the problem.

2.7.5 Consistency with the biological framing

Animals do not exchange kinematic specifications. Insects and vertebrates infer what another body is doing by observing it. If the claim under test is that behaviour is an abstraction separable from body shape, evidence gathered through the channel that biological imitation actually uses is more consistent with that claim than evidence gathered from privileged design parameters.

2.7.6 Costs accepted

Vision is noisier and more fragile than proprioception. Raw visual features are dominated by rendering style rather than behaviour, which forces strict control of camera, lighting, and background (Section 3.2.3), and two plausible background choices were found empirically to corrupt the signal in opposite ways. Pixels also widen the eventual sim-to-real gap relative to joint-space observation, and the frozen encoder is computationally heavy. These costs are accepted because Sections 2.7.1 and 2.7.2 make vision the setting in which the result, if obtained, actually means what the thesis claims it means.

CHAPTER 3 RESEARCH METHODOLOGY

3.1 System Overview

The system comprises four learned/parameterized components on top of a fixed perceptual front-end, trained end-to-end except for the frozen encoder:

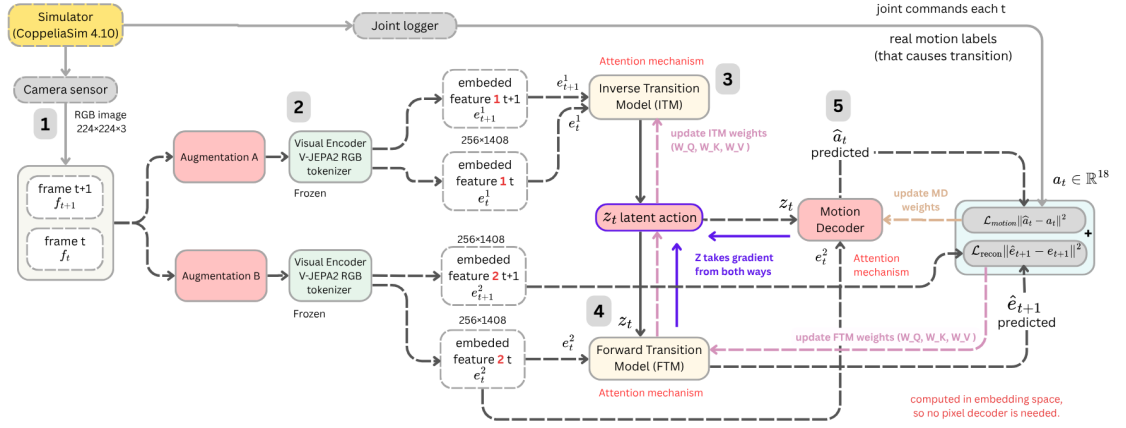


Figure 3.1 System pipeline. A frozen V-JEPA2 encoder maps each RGB frame to an embedding e_t ; the Inverse Transition Model infers the latent action z_t from (e_t, e_{t+1}) ; the Forward Transition Model predicts $\hat{e}_{t+1}$ from (e_t, z_t) (loss $\mathcal{L}_{\text{recon}}$); and the Motion Decoder maps z_t back to a body-specific joint command $\hat{a}_t$ (loss $\mathcal{L}_{\text{motion}}$).

Input/output specification of each block (units stated, per the advisors' requirement):

Table 3.1 Input/output specification of each block in the pipeline.

Block	Input	Output	Trained?
V-JEPA2 encoder	RGB frame, $256 \times 256 \times 3$ (uint8)	$e_t \in \mathbb{R}^{256 \times 1408}$ patch tokens	No (frozen)
Inverse Transition Model	$[e_t, e_{t+1}]$ (512 tokens)	$z_t \in \mathbb{R}^{64}$ (latent action)	Yes
Forward Transition Model	$[e_t, z_t]$	$\hat{e}_{t+1} \in \mathbb{R}^{1408}$	Yes
Motion Decoder	$(e_t$ as context, z_t as query)	$\hat{a}_t \in \mathbb{R}^{18}$ (joint targets, rad)	Yes (weights retained for downstream use)

Losses: $\mathcal{L} = \lambda_{\text{recon}} \mathcal{L}_{\text{recon}} + \lambda_{\text{motion}} \mathcal{L}_{\text{motion}}$, where $\mathcal{L}_{\text{recon}} = \|\hat{e}_{t+1} - e_{t+1}\|^2$ is computed in embedding space (no pixel decoder required) and $\mathcal{L}_{\text{motion}} = \|\hat{a}_t - a_t\|^2$. **Cross-augmentation** is applied: two independent augmentations of each frame pair are encoded, one feeding the ITM and the other the FTM target, which prevents the ITM from smuggling raw future-frame content into z_t as a shortcut.

3.2 Simulation Environment and Morphology Variants

3.2.1 Simulator and model

CoppeliaSim v4.10 (Bullet 2.78 engine, 20 Hz timestep). The robot is the *Medau-roidea extradentata* model (six legs, three actuated joints each: ThC/CTr/FTi), adopted from Larsen et al. (2023), with an 18-dimensional joint-position-target action space.

3.2.2 Leg-length variants

Three scenes are generated by scaling the coxa/femur/tibia segments of all six legs by a uniform factor and repositioning downstream joints accordingly: **long** 1.0× (**base**), **medium** 0.75×, **short** 0.5×. Scaling is verified numerically (measured foot-reach ratio matches the target factor for all six legs).

3.2.3 Observation capture

A single fixed vision sensor (256×256 , side view, $\sim 40^\circ$ elevation) is added programmatically so that its pose relative to the robot is *identical across all morphologies by construction*. The floor uses a matte, mildly-textured, non-repeating surface (a controlled choice: both a stark checkerboard and a blank surface were found to corrupt the visual features). The camera tracks the robot’s planar position at fixed height and orientation, keeping apparent size constant.

3.3 Visual Encoder (Frozen V-JEPA2)

The encoder is facebook/vjepa2-vitg-fpc64-256 (ViT-g/16, $\sim 1\text{B}$ parameters), used frozen. Because this checkpoint is natively a 64-frame *video* encoder with 2-frame tubelets, feeding a real clip would let each frame attend to other timesteps and contaminate its embedding. To obtain an independent per-frame embedding, each frame is **duplicated into the minimal 2-frame tubelet** and encoded alone, yielding 256 patch tokens of dimension 1408 per frame. This usage has direct precedent in V-JEPA 2-AC.

3.4 Latent Action Model

The two learned transition models below are instances of the inverse and forward dynamics models described in Section 2.3.1: the ITM infers the latent action from a state transition, and the FTM predicts the next state from the current state and that latent. This project refers to them as **Transition Models** rather than *dynamics* models deliberately. They operate entirely on **transitions between visual embeddings** produced by the frozen encoder. “Transition Model” names precisely what the module computes (a mapping over embedding transitions) and avoids implying that the model estimates the body’s physical dynamics, which it does not. The underlying technique is identical to the inverse/forward dynamics models of the literature; only the name is chosen for descriptive accuracy in the visual-embedding setting.

3.4.1 Inverse Transition Model (ITM)

Attention-based (causal self-attention blocks with a learned query token) mapping the pair $[e_t, e_{t+1}]$ to the latent action $z_t \in \mathbb{R}^{64}$. The causal structure frames z_t as “what happened between t and $t + 1$.” This is the inverse dynamics model of Section 2.3.1, computing z_t from a transition instead of a native action.

3.4.2 Forward Transition Model (FTM)

Attention-based, predicting the next visual embedding $\hat{e}_{t+1}$ from $[e_t, z_t]$; supervised by $\mathcal{L}_{\text{recon}}$ in embedding space.

3.4.3 Motion Decoder (MD)

Cross-attention with z_t as query over the current visual context, producing $\hat{a}_t$; supervised by $\mathcal{L}_{\text{motion}}$ against the auto-logged joint command a_t . This is the only place ground-truth action is used, and the decoder is **not part of the Phase 1 evaluation**, though its weights are retained. Its role during pretraining is to anchor z_t to real motion so the latent cannot collapse into a trivial identity. Beyond this thesis it has a second role: it is the only module that maps a latent action back into executable joint commands, so any downstream policy emitting z_t would require it. This also answers a standing question about the design, namely why a latent is introduced at all when the robot ultimately needs joint commands: the policy operates in the latent space because that is the part expected to transfer across bodies, while the decoder performs the body-specific conversion. The conversion separates what transfers from what does not.

3.4.4 Dimensions and hyperparameters

z_t is 64-dimensional (per LAC-WM §4.2). The λ weights, learning rate, and optimizer are to be determined by ablation (the base paper does not report them). Training uses mixed precision on a single GPU (RTX 2080 Ti, 11 GB), with small batch size and gradient accumulation to fit the frozen 1B-parameter encoder in the training loop (cross-augmentation prevents offline embedding caching).

3.5 Data Collection

3.5.1 Pilot dataset (reference-gait replay)

The preliminary results of Section 3.7 were obtained on a pilot dataset in which each morphology is driven by a single fixed reference gait (replayed real-animal joint trajectories from one *Medauroidea* individual), capturing per timestep the RGB frame, the 18-dimensional joint command a_t , the body pose, and the per-foot contact force. Reloading the scene between episodes provides genuine variation, because the contact-rich dynamics are chaotic and each reload yields a different trajectory, while within an episode the

simulation is deterministic. Behaviour is labelled by the **6-bit foot-contact state** (which of the six feet are planted), a body-pose label preferable to a time-based phase label for the reason given in Section 3.7.3.

3.5.2 Why the pilot dataset is not sufficient for the main experiment

In the pilot, the *same* joint-command sequence is applied to all three bodies (this is exactly what makes the morphology-gap test of Section 3.7.1 valid: identical input, different outcome). Verification confirms the commands are bit-identical across bodies (maximum pairwise difference is zero to machine precision). This property, correct for the gap test, is fatal for training the latent action model. The Motion Decoder is grounded by $\mathcal{L}_{\text{motion}} = \|\hat{a}_t - a_t\|^2$; if a_t is identical across bodies, the target carries no morphology information, the decoder has no reason to condition on the body, and there is nothing to *retarget*. The latent action z_t would then be describable as a body-independent action simply because the *command* was already body-independent, which is a vacuous form of the property this thesis claims to demonstrate. The main experiment therefore requires a dataset in which the *behaviour* is shared across bodies but the *command* differs per body.

3.5.3 Dataset redesign (inverse-kinematics retargeting)

The redesign defines each behaviour not as a joint trajectory but as a **task-space foot trajectory** (a Cartesian path for each foot). Given a shared foot trajectory, inverse kinematics is solved *separately for each morphology* to obtain the joint commands that place that body’s feet on the shared path: $a_t^{\text{body}} = \text{IK}(\text{foot trajectory}, \text{body})$. Because foot position depends on both joint angle and link length, the same foot target yields different joint angles on a long leg and a short leg, so the resulting commands differ per body while the intended behaviour is held constant. This is the precise condition the Motion Decoder must learn to recover: a body-independent intent (the latent action) mapped to body-specific joint commands. The source of the shared foot trajectories is a mature logged expert dataset from Larsen et al. (2023) (`expert_66k_aug3c_fcontact`, 66,000 timesteps) that records foot trajectories and the simulator’s own binary contact labels directly, so the trajectories need not be re-derived and the contact labels do not depend on a hand-chosen force threshold. IK is a closed-form solve requiring no additional training, and it is introduced as an *addition* to the pipeline rather than a change to the model architecture. A single successful IK pipeline also supplies the multiple behaviours (walk, turn, and stop) as additional foot trajectories, which the latent-space validation of Section 3.6.2 requires.

[LIMITATION] *IK produces comparable task-space objectives but does not guarantee identical contact dynamics across morphologies, and the anticipated objection that “learning the Motion Decoder is merely learning IK” is answered by the claim under test, namely that the model recovers this body-specific retargeting from observation alone, without being given the kinematics.*

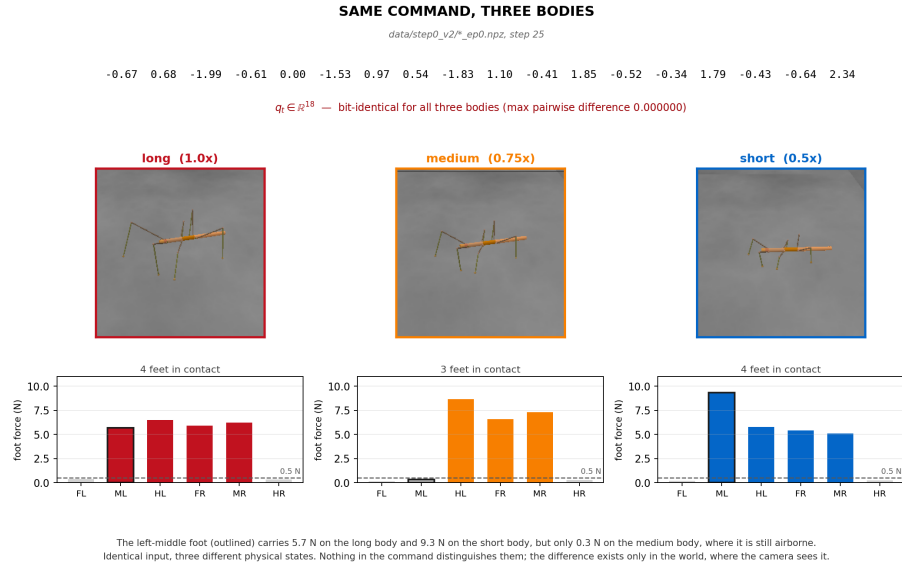


Figure 3.2 A single timestep with a bit-identical joint command across the three bodies. The left-middle foot carries 5.7 N on the long body and 9.3 N on the short body but only 0.3 N on the medium body, where it is still airborne. The command cannot distinguish the bodies; the difference exists only in the resulting physical state.

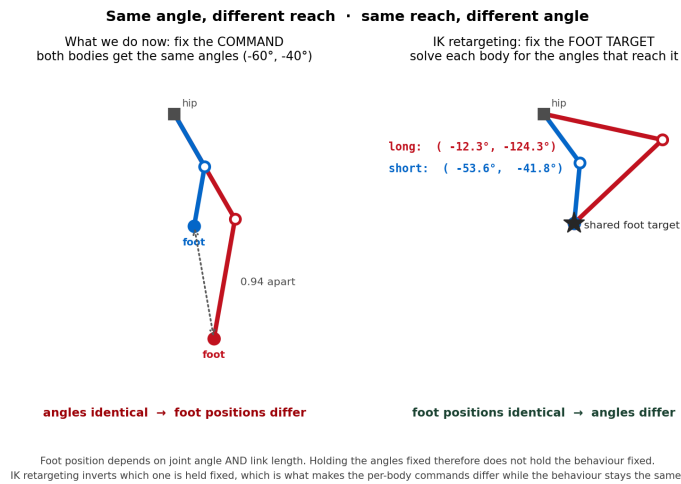


Figure 3.3 Why per-body commands are needed. Left: identical joint angles place the foot in different positions on a long versus a short leg. Right: inverse kinematics inverts the constraint, so a shared foot target is reached by different joint angles per body, which is the body-specific command the Motion Decoder must learn to produce.

3.6 Evaluation Protocol

3.6.1 Metrics

Two complementary measures are reported for every latent-structure claim, since reporting only one can mislead. The first is **linear-probe accuracy**: can a simple classifier decode a label (behaviour or morphology) from the representation? This measures the *presence* of information. The second is the **silhouette score together with between-class variance**, which measure *dominance*, that is, whether that information is the main axis of variation. Cross-morphology **transfer** is measured by training a probe on two morphologies and testing on the held-out one.

3.6.2 Milestone gates

- *Morphology gap*: pass if the three variants travel measurably different distances under identical commands.
- *Encoder sanity*: pass if behaviour (foot-contact state) is decodable from e_t above a noise floor.
- *Latent validation*: pass if z_t lowers morphology decodability and raises cross-morphology behaviour transfer relative to the raw encoder baseline.
- *Transfer*: pass if the pretrained world model reaches the same $\mathcal{L}_{\text{recon}}$ on the unseen morphology with significantly fewer episodes than a from-scratch model.

3.6.3 Primary evidence and the value-over-raw-command check

The **primary evidence** for the thesis is the latent-space validation: does z_t raise cross-morphology behaviour transfer and lower morphology decodability relative to the raw encoder e_t , both at once. This test is clean because it examines z_t on its own: it needs only the held-out body’s video, and it does not compare against the native command.

Does the latent add value over the raw command? This is answered by two measures. The value of the latent is therefore assessed by:

1. **Adaptation efficiency (the primary usefulness metric, per the lab advisor).** Starting from the pretrained world model, adapt to the held-out body and measure the number of training episodes needed to reach a target reconstruction error, against a **from-scratch** model and against a model **pretrained with the raw command** instead of the latent. This measures *how quickly* the representation transfers, which is fair because every model is actually trained/adapted, and it is the metric the lab advisor identified as the main test of whether the world model is useful. The observation-only start (no action) is included to isolate what the action representation contributes.

2. **The availability argument.** The raw command a_t is a *privileged* input: obtaining the correct command for a new body requires that body’s kinematics (through the inverse-kinematics retargeting of Section 3.5.3), which the motivating scenario assumes is unavailable. The latent z_t is inferred from observation and needs no kinematics. A latent that *matches* what the privileged command achieves is therefore already a substantive result, because it recovers from vision alone what otherwise requires the body’s internal description.

A further contrast, where applicable, is an explicit morphology-conditioned model in the spirit of QWM, comparing explicit and implicit morphology handling.

3.7 Preliminary Validation of the Method

The first two milestones of Section 1.5 have been executed on a **pilot dataset** (the reference-gait replay of Section 3.5), and their outcomes justify the design choices above. These are diagnostic results whose purpose is to establish that the methodology is sound before the main training begins; they are **not** the thesis’s final measurements. They will be regenerated on the redesigned IK dataset (Section 3.5), and the exact values are expected to shift. Every figure below is reproducible from the recorded data. Full results belong in the results chapter of the completed thesis.

3.7.1 The morphology gap is measurable and statistically decisive

All three variants were driven with a bit-identical 18-dimensional command sequence, and the body’s world-frame position was read directly from the simulator. Two distance measures are reported: *path length* (the distance actually walked) and *net displacement* (straight-line progress), because the open-loop gait has no heading correction and its side-to-side body oscillation must not be mistaken for a locomotion difference.

Table 3.2 Distance travelled under an identical command sequence, five episodes per morphology.

Morphology	Path length ($n = 5$)	Net displacement ($n = 5$)
Long, 1.0×	5.217 ± 0.069 m	4.404 ± 0.187 m
Medium, 0.75×	4.149 ± 0.019 m	3.569 ± 0.010 m
Short, 0.5×	3.228 ± 0.011 m	2.729 ± 0.011 m

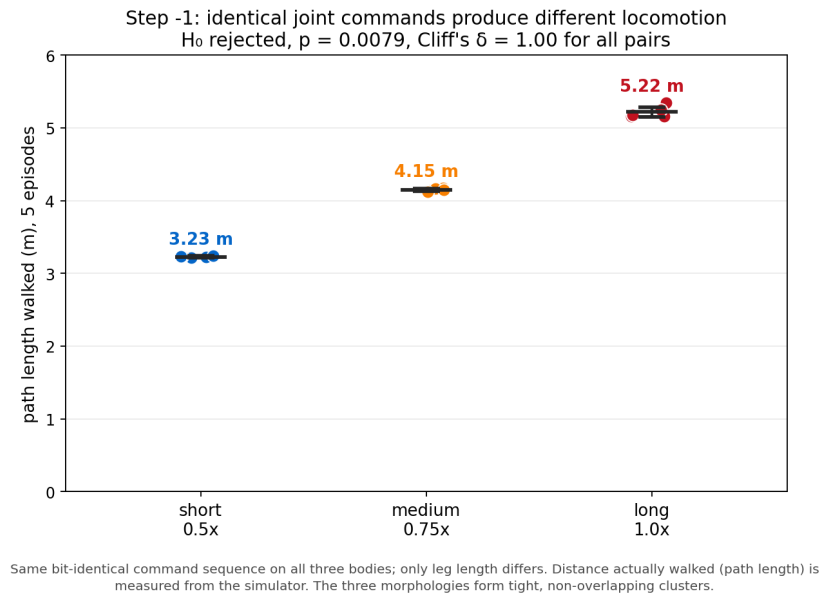


Figure 3.4 Left: top-down body trajectories (five episodes per morphology) under an identical command sequence, aligned to a common start and heading and drawn at true aspect ratio. Right: per-episode path length and net displacement with mean $\pm$ standard deviation; the three morphologies do not overlap.

A two-sided Mann-Whitney U test rejects the null hypothesis of equal outcomes for every pair of morphologies at $p = 0.0079$, with Cliff's $\delta = 1.00$ (complete separation: every episode of a longer-legged body exceeds every episode of the shorter one; the worst long-leg net displacement, 4.032 m, still exceeds the best short-leg value, 2.746 m). At $n = 5$ per group, $p = 0.0079$ is the smallest value the test can return, i.e. the strongest statement this sample size permits. The relationship is not a simple rescaling: walking speed falls with roughly the 0.69 power of leg length rather than the 1.0 that naive geometric scaling would predict. This confirms that identical commands yield genuinely different motion, which is the premise of the whole study.

3.7.2 The frozen encoder carries the required behaviour signal, entangled with body shape

Using a linear probe (logistic regression with standardisation) on the mean-pooled 1408-dimensional embedding, foot-contact state (which feet are planted, top-8 patterns) is decodable **within a body** at a mean macro-F1 of 0.84 (long 0.83, medium 0.95, short 0.78; per-body variance is high, ± 0.18 on the long body, because only five episodes are collected), against a shuffled-label control at chance (macro-F1 ≈ 0.12 , chance = 0.125). The Inverse Transition Model therefore has a usable behaviour signal to extract. The same probe **across bodies** (train on one morphology, test on another) collapses to a mean macro-F1 of 0.16: the behaviour information is present but **entangled with body shape**, and transfer degrades as the leg-length gap widens. Morphology itself is separately decodable from the raw embedding at approximately 100 percent (holding under episode-grouped cross-validation, so it is not frame memorisation), and its leading principal component

orders the three bodies by leg length with no labels supplied. This is the expected and desirable baseline: the encoder plainly sees that the bodies differ, and removing that difference while keeping the behaviour signal is precisely the job of the latent action model. **Two caveats define the pilot status of these numbers:** each morphology was recorded in a single session, so morphology and recording session (lighting, background) are perfectly confounded and cannot be separated on this data; and the top-8 contact patterns cover only 43 percent of frames because the replayed gait is a phase-staggered wave rather than a clean tripod, so the within-body figure is measured on the cleaner, more frequent half of the data and is likely optimistic. The multi-session control needed to attribute the morphology signal to leg length rather than session is scheduled with the dataset redesign.

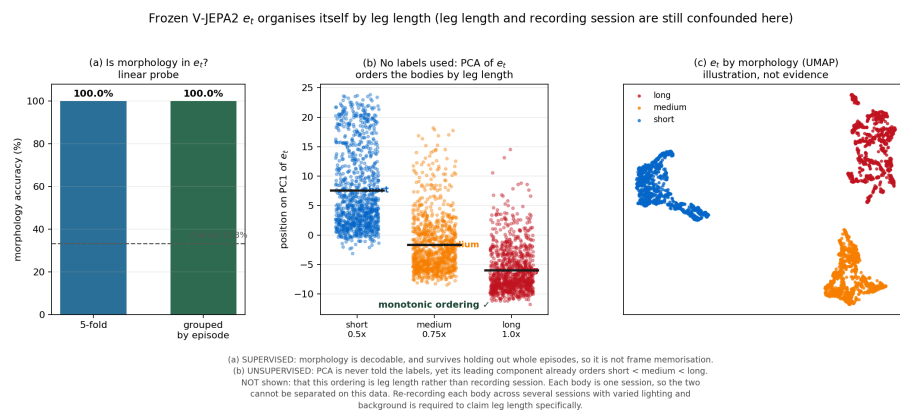


Figure 3.5 Morphology is strongly and ordinally encoded in the frozen embedding. (a) A supervised linear probe reaches ~100 percent and holds under episode-grouped cross-validation. (b) Unsupervised PCA, given no labels, orders the three bodies by leg length along its leading component. (c) A UMAP projection, shown for illustration only. Leg length and recording session are confounded in this pilot.

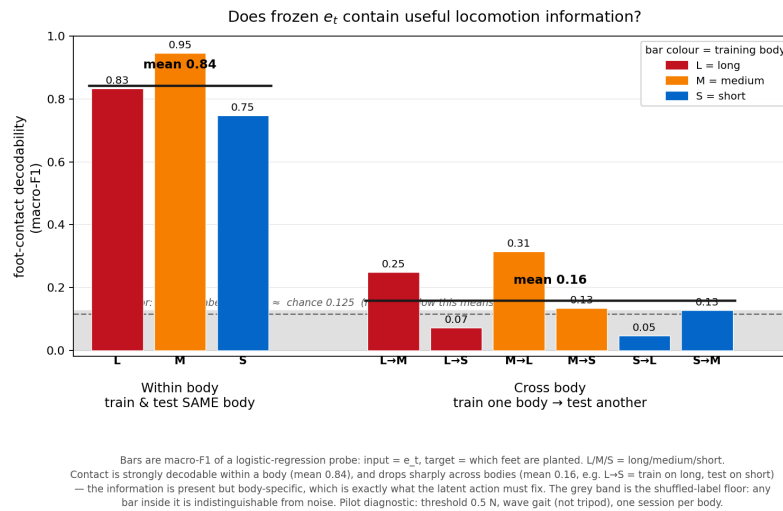


Figure 3.6 Foot-contact behaviour is decodable within a body (mean macro-F1 0.84) but collapses across bodies (mean 0.16), and a shuffled-label control sits at the chance floor. The behaviour information is present but entangled with body shape, the signal the latent action must preserve while making it body-independent.

3.7.3 Two methodological findings that shaped the protocol

First, a representation can *contain* a signal without that signal *dominating* it, so a single metric can mislead. Behaviour is decodable from the raw embedding (probe macro-F1 0.84 within a body), yet morphology is the dominant axis of variation (morphology silhouette +0.0835 against a much weaker behaviour signal, with morphology decodable at ~100 percent). A linear probe measures presence; the silhouette score and between-class variance measure dominance. Section 3.6.1 therefore requires both, and additionally reports balanced accuracy / macro-F1 because the contact-pattern classes are strongly imbalanced. Second, an early time-based behaviour label (step index modulo a hand-chosen gait length) was abandoned: identical commands do not place different bodies in the same pose, so a time index is not a body-independent behaviour label, and it was found to be an artefact of the chosen trim window rather than a physical gait state. A pose-based label (foot-contact state, read from force sensors) is used from this point forward, and Section 3.5 records its residual limitations.

3.8 Remaining Work and Next Steps

The two preliminary milestones of Section 3.7 establish that the morphology gap is real and that the frozen encoder carries a usable, morphology-entangled behaviour signal. This section states, concretely and in order, every step that remains, so that the scope of the committed work is explicit. Each step names what it produces and the condition under which it is considered complete.

Step A: Dataset redesign (inverse-kinematics retargeting). Build the IK pipeline of Section 3.5.3: take the shared task-space foot trajectories from the mature expert log of Larsen et al. (2023) (`expert_66k_aug3c_fcontact`), solve `simIK` separately per morphology to obtain per-body joint commands, and drive each body with its own command while recording, per timestep, the RGB frame, the joint command a_t , the body pose, and the simulator’s binary foot-contact labels. *Produces:* a dataset in which behaviour is shared across bodies but the command differs per body. *Gate:* a_t is verified to differ across bodies (variance well above machine precision), removing the pilot’s shared-command limitation of Section 3.5.2.

- *Sub-step A1 (confound control).* Record each morphology across **several sessions** with varied lighting and background, so that a probe trained on one session can be tested on another. This is the control needed to attribute the morphology signal of Section 3.7.2 to leg length rather than to recording session.
- *Sub-step A2 (behaviours).* Generate the walk, turn, and stop trajectories as task-space foot paths (Section 1.4); the latent validation of Step D uses more than one behaviour.

Step B: Regenerate the preliminary analyses on the redesigned dataset. Re-run the morphology-gap and encoder-sanity analyses of Section 3.7 on the IK dataset, using the corrected evaluation of Section 3.6.1 (macro-F1 / balanced accuracy for the imbalanced contact classes, episode-grouped cross-validation, and the multi-session probe from A1). *Produces:* the final baseline values for e_t against which z_t will be compared. *Gate:* the qualitative conclusions of Section 3.7 hold on the clean dataset; the morphology-versus-session attribution is resolved.

Step C: Train the latent action model (ITM + FTM + Motion Decoder). Train the three modules of Section 3.4 jointly with the combined loss and cross-augmentation, on the two training morphologies (short and long). *Produces:* the learned latent action z_t and the trained transition and decoding modules. *Gate:* both losses converge and z_t does not collapse (the Motion Decoder reconstruction stays informative).

Step D: Latent-space validation (the two-sided criterion). Probe z_t for behaviour transfer and for morphology decodability, and cluster z_t by behaviour across bodies. *Gate:* z_t lowers morphology decodability relative to e_t **and** raises cross-morphology behaviour transfer relative to e_t , measured together (Sections 2.6.4, 3.6.1).

Step E: The decisive ablation (run early, not last). Compare a latent-conditioned forward model, $F(e_t, z_t)$, against a raw-joint-conditioned one, $F(e_t, a_t)$, and an observation-only one, $F(e_t, 0)$, under an identical encoder, capacity, dataset, and budget (Sections 2.6.3, 3.6.3). *Gate:* whether the latent improves held-out prediction over the raw command is the central question; this step is scheduled before the transfer experiment so that a negative answer is discovered early.

Step F: Transfer to the unseen morphology. Adapt the pretrained model to the held-out medium body and measure the number of episodes needed to reach a target reconstruction error, against a from-scratch baseline. *Gate:* the pretrained model reaches the target with significantly fewer episodes (the sample-efficiency claim of Section 3.6.3), or the result is reported as negative.

Step G: Analysis, writing, and scoped extensions. Consolidate results. Extensions that the pipeline makes cheap but that are not part of the core claim: an out-of-range (extrapolation) morphology beyond the $0.5\times$ to $1.0\times$ training range, and an explicit cross-embodiment alignment objective in the manner of UniSkill (Section 2.3.2). These are stated as directions, not commitments.

3.9 Work Plan

The timeline follows the steps of Section 3.8. The simulation setup and both preliminary milestones (Section 3.7) are complete; the remaining steps span August to November, with the decisive ablation (Step E) scheduled early so a negative answer would surface before the transfer experiment.

Table 3.3 Work plan. The core experiments (through Step F) target completion by the end of October, leaving November for analysis and writing.

Step	Activity	Aug	Sep	Oct	Nov
(setup)	Simulation setup, morphology variants, vision capture, preliminary milestones (§3.7)	done			
A	Dataset redesign: IK retargeting, multi-session capture (A1), behaviours (A2)	✓			
B	Regenerate the preliminary analyses on the IK dataset (final e_t baselines)	✓	✓		
C	Train the latent action model (ITM + FTM + Motion Decoder)		✓		
E	Decisive ablation: latent vs raw-command vs observation-only (run early)		✓		
D	Latent-space validation (two-sided: behaviour up, morphology down)		✓	✓	
F	Transfer to the held-out medium body; sample-efficiency vs from-scratch			✓	
G	Analysis, thesis writing, scoped extensions			✓	✓

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